

VCD-2703/5-1-FYCS-SEMESTER I-75 Marks-2 ½ Hours

Instructions: All Questions are compulsory.

Sub Questions from a-h each carry five marks.

Q1. Answer any Four Questions. 20

- Describe block diagram of computer?
- Explain history of computer?
- Classify the computer based on its speed and technology used?
- Define information and give its characteristics?
- Differentiate between data and information?
- Define data and give methods of collecting various forms of data?
- Convert binary to decimal
A) $(1111111)_2$ B) $(101101.10101)_2$
- Convert hexadecimal to decimal
A) $(3A.2F)_{16}$ B) $(2A3.A)_{16}$

Q2. Answer any Four Questions. 20

- Explain AND gate and OR gate.
- Explain NAND gate and NOR gate.
- Explain EX gate and NOR gate.
- Explain S-R flip flop.
- Explain J-K flip flop.
- Explain full adder with circuit diagram.
- Explain half adder with circuit diagram.
- Explain multiplexer with circuit diagram.

Q3. Answer any Four Questions. 20

- Differentiate between SRAM and DRAM.
- Explain ROM with its types.
- Explain the concept of virtual memory.
- Explain the concept of cache memory with its need.
- What is an accumulator?
- Explain characteristics of multiprocessor?
- Explain 8085 microprocessor architecture?
- Explain different addressing modes?

Q4. Answer any Three Questions.

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- a) Explain excess -3 code and gray code
 - b) Binary multiplication
A) $(1001)_2 \times (1101)_2$ B) $(1111)_2 \times (1101)_2$
 - c) Convert decimal to octal
A) $(3287.510098)_{10}$ B) $(0.6875)_{10}$
 - d) Explain encoder with circuit diagram.
 - e) Explain decoder with circuit diagram.
 - f) Explain binary adder/subtractor.
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